MILITARY SPECIFICATION
PRIMER, PERCUSSION: CCU-20/A AND PRIMER, PERCUSSION: CCU-21/A

1. SCOPE

1.1 This specification covers the CCU-20/A and CCU-21/A percussion primers used in cartridge actuated devices and other applications where high temperature stable primers are required.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

C11738784	Primer, Percussion, CCU-21/A
C11738777	Primer, Percussion, CCU-20/A
D10524200	Case, Cartridge, 5.56mm
C11738782	Primer, Composition FA988
F10535777	Packing, and Marking for Primers, Small Caliber Ammunition Cartons; (Plastic Tray) Box, Ammunition M2A1 Box Wirebound
B11738781	Pellet
IEL-C11738784	Index of Inspection Equipment List for Primer, Percussion, CCU-21/A
IEL-C11738777	Index of Inspection Equipment Lists for Primer, Percussion, CCU-20/A

PUBLICATIONS

ORD-SIP-S314	Visual Inspection Standards for Small Arms Ammunition Primer Defects
AMSMU-P 715-501FA1	Ammunition Ballistic Acceptance Test Methods-Test Procedures for 5.56mm Cartridge

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.- Materials and components shall be in accordance with applicable drawings and specifications.

3.2 Primer assembly.- The CCU-20/A and CCU-21/A primers shall comply with the requirements of Drawing C11738777 and C11738784, respectively.

3.3 Sensitivity.- The sensitivity shall fall within the limits specified as follows for each type of primer:

Primer	Required Case	Height in Inches H+3	H-2
CCU-20/A	5.56mm	20	3
CCU-21/A	5.56mm	18	1

3.4 Priming composition.- The priming composition for the two primers shall comply with the requirements of Drawing C1178782.

3.5 Pellet weight.- The pellet weights of the CCU-20/A and CCU-21/A primers shall comply with the requirements of Drawings B11738781 and C11738784, respectively.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings and as follows:

3.6.1 Metal defects.- The primer shall be free of cracks, laminations, nicks, dents, scratches and other metal defects.

3.6.2 Foreign matter.- The primer shall be free of corrosion, stains, discoloration, dirt, oil and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 1000 primers. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only, and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of primers shall be identified as to type, quantity, lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.15 percent and the AQL for the minor shall be 1.50 percent.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

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TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Primer				
1.	No anvil	X			
2.	Double anvil	X			
3.	No charge	X			
4.	No foil		X		
5.	Double of multiple foil		X		
6.	Slipped foil				X
7.	Inverted or sidewise inserted anvil	X			
8.	Crushed or multilated				X
9.	Loose composition				X
10.	Deep seated anvil		X		
11.	Cocked anvil				X
12.	Low anvil		X		
13.	High anvil				X
14.	Excessive lacquer			X	
15.	Loose anvil		X		
16.	Stained or corroded metallic components				X
17.	Foreign matter		X		
	Cup				
18.	Bell shaped		X		
19.	Bad bevel			X	
20.	Brass shavings				X
21.	Cracked				X
22.	Crooked or eared				X
23.	Dents				X
25.	Crown on flat cup				X
27.	Laminated metal				X
29.	Scratched				X
30.	Nicked				X
31.	V or U				X
32.	Low cup			X	
	Anvil				
33.	Burred			X	
34.	Chipped				X
35.	Filled		X		
36.	Partial	X			
37.	No vent holes	X			
38.	Laminated metal		X		
39.	Ringed			X	
40.	Unsymetrical vent holes			X	
	Foil				
41.	Partial			X	
42.	Stained	X			

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TABLE I
(Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Pellet				
43.	Heavy		X		
44.	Light	X			
	Gaging Primer				
45.	Diameter		X		
46.	Excess tolerance between long and short axis diameter		X		
47.	Height			X	

1/ Refer to ORD-SIP-S314 for visual defect standards for defects 1 through 44.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. The primers shall be selected in such a manner that the sample is representative of the entire lot. Only primers conforming to the visual and dimensional requirements shall be subjected to the primer sensitivity test.

TABLE II

Tests	No. of Primers at ambient Temperature	Requirement Paragraph
Sensitivity 1/	600	3.3
Priming Composition 2/	*	3.4
Pellet weight 3/	50	3.5

*Number of primers as are necessary to yield 3 grams of priming composition.

1/ The test sample shall consist of the number of primers required for performance of a complete run down test wherein fifty primers are tested at each height. Failure of the primers to comply with the applicable requirements shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity used in the first test. Failure of the primers of the second sample to comply with the applicable requirements shall be cause for rejection of the lot.

2/ Failure of the priming composition to comply with the applicable requirements shall be cause for rejection of the primer lot.

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3/ Failure of the pellets to comply with the applicable weight requirements shall be cause for rejection of the lot subject to testing of two additional samples of fifty primers each. Failure of the pellets in either or both of the additional samples to comply with the applicable weight requirements shall be cause for rejection of the lot.

4.3.4 Packing and marking examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table III, as applicable, using inspection level 1 of MIL-STD-105. The AQL for the major class shall be 0.65 percent and the AQL for the minor class shall be 2.5 percent.

4.3.4.1 Classification of defects.- The classification of packing and marking defects shall be as specified in Table III.

TABLE III

No.	Defect	Critical	Major	Minor
	Carton, before sealing			
1.	Improper quantity of primers		X	
2.	Improper positioning of primers			X
	Carton, after sealing			
3.	Label missing or insecurely applied		X	
4.	Carton torn or ripped		X	
5.	Marking missing, incorrect or illegible		X	
	Ammunition box, before closing			
6.	Contents, incorrect			X
7.	Filler(s) missing or improper			X
	Ammunition box, after closing			
8.	severe dents or buckles			X
9.	Incorrectly closed and secured box		X	
10.	Marking missing, incorrect or illegible			X
	Wirebound box, before closing			
11.	Filler or separator missing			X
	Wirebound box, after closing			
12.	"Handle Carefully" marking missing incorrect or illegible	X		
13.	Other marking missing, incorrect or illegible			X
14.	Unfastened binding wire			X
15.	Missing box end		X	
16.	Missing or incorrectly attached car seal		X	

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4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-C11768784 or C11738777 as applicable.

4.4 Test methods and procedures.

4.4.1 Sensitivity test.- The primer shall be inserted into the primer pocket of the 5.56mm cartridge case so that the surface of the primer cup, when measured from the center of the primer, is within .000 to .003 inch below flush of the cartridge case head provided the pocket depth is .118 + .004 inch. Prior to insertion of the primer into the primer pocket of the cartridge case a .005 inch thick aluminum or magnesium disc (or other suitable material approved by the contracting officer) shall be placed in the pocket to cover the bottom of the pocket. The test shall be conducted in accordance with complete run-down test procedures of AMSMU-P 715-501FA1 except that calculations for skewness factor ($^{a}3$) are not required.

4.4.1.1 Second sample.- Second sample tests for sensitivity shall be made by testing 100 primers at each height.

4.4.2 Priming composition analysis.

4.4.2.1 Potassium chlorate.

4.4.2.1.1 Reagents.

4.4.2.1.1.1 Ferrous ammonium sulfate solution (0.1 N).- Dissolve 40 gm of $Fe(NH_4)_2SO_4.6H_2O$ in about 500 ml of sulfuric acid (1 to 3) and dilute to 1 liter with sulfuric acid (1 to 3).

4.4.2.1.1.2 Potassium permanganate solution (0.1 N).

4.4.2.1.2 Procedure:

Transfer approximately 0.15 gm of sample to a tared 15-ml sintered glass crucible of fine porosity, weigh the crucible plus sample, and calculate the exact weight of the sample by difference. Extract with a total of 50 ml of hot water (temperature almost 70°C) added in about 5-ml portions and collect the filtrate in a 250-ml beaker under a Witt filter apparatus. In the first three extractions, allow to stand without section with occasional stirring for 5 minutes. Save the crucible and residue for the determination of antimony sulfide and calcium silicide. To the filtrate add 50 ml of sulfuric acid (1 to 1) and cool to room temperature. Add 50.00 ml of 0.1 N ferrous ammonium sulfate solution with a pipet and allow to stand for 5 minutes. Titrate with 0.1 N potassium permanganate solution to a pink color that persists for 30 seconds. Determine a blank titration by pipetting 50.00 ml of 0.1 N ferrous ammonium sulfate solution into a mixture of 50 ml of water and 50 ml of sulfuric acid (1 to 1) and titrating with the permanganate.

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Calculate the percent potassium chlorate as follows:

$$\text{Percent Potassium Chlorate} = \frac{2.043\ (A-B)N}{W}$$

where, A - ml of potassium permanganate solution for blank titration
B - ml of potassium permanganate solution for sample titration
N - normality of potassium permanganate solution
W - gm of sample

4.4.2.2 Antimony sulfide.

4.4.2.2.1 Reagents.

4.4.2.2.1.1 Potassium bromate solution (0.05N).- Dissolve 1.3917 gm of potassium bromate in water and dilute to 1 liter in a volumetric flask. This is a primary standard.

4.4.2.2.1.2 Methyl orange indicator.- Dissolve 0.1 gm of methyl orange in 100 ml of water.

4.4.2.2.2 Procedure.

Transfer the sintered glass crucible (and residue) from the potassium chlorate determination to a wide-mouth 500-ml Erlenmeyer flask and add 15 ml of concentrated nitric acid and 10 ml of concentrated sulfuric acid. Carry along a reagent blank. Cover the flask with a watch glass and evaporate to fumes of sulfuric acid. Add nitric acid dropwise and continue fuming until the organic matter is destroyed. Allow to cool, wash down glass and sides of the flask, and again evaporate to fumes of sulfuric acid (without the watch glass). Do not remove the crucible. Allow to cool and add 20 ml of water, 75 ml of concentrated hydrochloric acid, and 1 ml of bromine. Boil without a watch glass until the yellow color due to the bromine disappears. Add 1 gm of sodium sulfite and wash down the dies. Boil for 12 to 15 minutes without a watch glass, adding boiling water about midway during this boiling period so as to maintain the volume at about 90 to 110 ml. Dilute to about 350 ml with boiling water and titrate with 0.05N potassium bromate solution, using methyl orange indicator (4 drops), preferably added toward the end of the titration. The end point is the disappearance of the methyl orange color. Save the solution and crucible for the determination of calcium silicide.

Calculate the percent antimony sulfide as follows:

$$\text{Percent antimony sulfide} = \frac{8.49\ (A-B)N}{W}$$

where, A - ml of potassium bormate solution for sample titration
B - ml of potassium bromate solution for blank titration
N - normality of potassium bromate solution
W - gm of sample

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4.4.2.3 Calcium silicide.

4.4.2.3.1 Reagents.

4.4.2.3.1.1 Ammonium oxalate solution (5%).- Dissolve 50 gm of ammonium oxalate in hot water, cool, and dilute to 1 liter.

4.4.2.3.1.2 Ammonium oxalate solution (1%).- Dilute 100 ml of ammonium oxalate solution (5%) to 500 ml.

4.4.2.3.2 Procedure.

Remove the crucible from the solution from the antimony determination, wash it well with water, and wash the solution and residue from the flask into a 600-ml beaker. Evaporate to fumes of sulfuric acid. Allow to cool, add 15 ml of hydrobromic acid (48%), and again evaporate to fumes of sulfuric acid (without a watch glass). Allow to cool and wash the solution and residue into a large platinum dish. Add 15 ml of concentrated nitric acid and 2 ml of hydrofluoric acid (48%) and evaporate to complete dryness. Add 20 ml of hydrochloric acid (1 to 1), warm, wash into a 660-ml beaker with water, and boil for a few minutes. Dilute to about 200 ml, heat to boiling, add 50 ml of ammonium oxalate solution (5%), neutralize to litmus paper with ammonium hydroxide and add 1 ml excess ammonium hydroxide. Digest at about 90°C for 1 to 2 hours, Filter through a Whatman No. 40 filter paper, transfer, and wash with ammonium oxalate solution (1%). Place the filter paper and percipitate into a tared platinum crucible, char, and burn off the filter paper at dull red heat. Ignite over a blast burner for 15 minutes, cool in a disiccator, and weigh quickly.

Calculate the percent calcium silicide as follows:

$$\text{Percent calcium silicide} = \frac{2.15C}{W}$$

where, C - gm of residue
W - gm of sample

Note: The factor 2.15 is an empirical factor obtained using commercial calcium silicide of the type employed in primers.

4.4.3 Pellet weight.- Each primer shall be weighed. The anvil disc and all priming mixture shall then be removed from the primer cup. The primer cup, anvil and disc shall be weighed together. The weight of primer pellet is: $W=A - B$

Where W - weight of primer pellet, grains
A - weight of assembled primer, grains
B - weight of cup, anvil, and disc, grains

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Levels of packing A and C shall comply with FED-STD-102

5.1.1.1 Level A.- The primers shall be packed in accordance with Drawing F10535777 or as required by contract.

5.1.1.2 Level C.- Unless otherwise specified by the procuring agency, the primers shall be packed in standard commercial containers so as to insure acceptance by common or other carrier for safe transportation to the point of delivery. The primers shall be packed and spaced in such a manner that a barrier of suitable material is present between the crown of one primer and the base of another primer in order to resist mass detonation and shield the primers from contact with stray electrical current, flame and shock.

5.2 Marking.

5.2.1 Level A.- Packing boxes shall be marked and labeled in accordance with the drawing cited in 5.1.1.1.

5.2.2 Level C.- Packing boxes shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

a. Title, number and date of this specification
b. Applicable Primer Percussion:
c. Applicable packing container (see 5.1.1)
d. The appropriate cartridge cases and test equipment when necessary for test purposes shall be supplied by the contracting officer.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1377-A724

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Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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